

A Branchless Calculus of Decision under Latency

Data-Oblivious Computation, the Geometry of the Latency–Quality Manifold,
and a Latency-Parameterized Theory of Decision-Engine Skill

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Abstract

We develop, from first principles, a theory of *decision under a hard latency budget*. The orthodox engineering instinct optimizes *depth*: the quality of a decision given time. We argue this is the wrong objective whenever the decision deadline is short, and we make the argument rigorous along three mathematical axes. **Algebra:** we show that data-dependent control flow is an avoidable luxury by constructing the *mask algebra* $(\mathbb{B}^w, \wedge, \vee, \neg, \oplus, \text{sel})$ and a predicate-mask homomorphism φ , proving that any bounded decision procedure compiles to a single-path (cyclomatic-complexity one) program. **Calculus:** we define the *latency functional* $\mathcal{L}$ and prove that data-oblivious programs have zero latency variance and a tight worst-case execution time, while branchy programs have latency that increases with input entropy. **Geometry:** we model engines as curves on the latency-quality plane and prove the *Inversion Theorem*: the quality gap between an adaptive search engine and a constant-time branchless engine is monotone in the budget and the search engine has a hard left feasibility boundary $\tau_{\min}$ below which it cannot act at all, leaving the branchless engine as the unique occupant of the sub- $\tau_{\min}$ regime. We instrument the theory with a chess engine—used purely as a measurement apparatus—built in a single afternoon, and report a 104-game latency sweep against Stockfish constrained to its minimum search. The data realize the predicted $E(\tau)$ curve, including parity with full-effort minimal-time Stockfish at 10 ms and coherent play at $3\mu\text{s}$ using a quarter of the opponent’s total compute. We close by defining *Decision-Engine Elo*, a latency-parameterized skill rating that does not currently exist for trading engines, and project the entire apparatus onto ultra-high-frequency trading, where the relevant quantity is not skill but skill-per-microsecond and where constant-time decisions admit a *provable* worst-case latency that no branchy system can certify.

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Chapter 1

Introduction

1.1 The thesis in one sentence

Below a critical latency, the optimal decision architecture is not the one that computes the best action but the one whose cost does not depend on the data—and the crossover is provable, measurable, and economically decisive.

1.2 The orthodoxy and its blind spot

The dominant paradigm of high-quality automated decision-making—epitomized by the chess engine **Stockfish**, the product of more than a decade and a half of adversarial search engineering—spends time to buy quality. Search depth d grows logarithmically in allotted time τ , and strength grows roughly linearly in depth. This is a magnificent strategy when time is abundant. It is a *liability* when it is not. In ultra-high-frequency trading (UHFT) the decision deadline is on the order of microseconds; a “grandmaster” that needs a full second has, in market terms, already missed the trade. The blind spot of the orthodoxy is that it optimizes a variable—depth—that is unavailable in the regime that matters.

1.3 Contributions

1. A self-contained **branchless algebra** (Chapter 2) and a compilation theorem turning bounded control flow into single-path arithmetic.
2. A **calculus of data-oblivious latency** (Chapter 3): zero-variance latency, tight WCET, and an entropy lower bound on branchy cost.
3. The **geometry of the latency–quality manifold** (Chapter 4) and the *Inversion Theorem*.
4. A **branchless decision engine** (Chapter 5): an affine evaluator with the split-ReLU signed identity, divergence-free batched evaluation, and two search operators (alpha–beta and Monte-Carlo tree search).
5. A **branchless process-conformance** layer (Chapter 6) giving every decision a certifiable, auditable execution receipt.

6. An **empirical instrument** (Chapter 7): chess as a zero-sum measurement apparatus, with a 104-game $E(\tau)$ sweep.
7. **Decision-Engine Elo** (Chapter 8), a latency-parameterized skill rating, and its UHFT projection (Chapter 9).

1.4 A note on method

Chess is not the object of study; it is the *micrometer*. It supplies a clean, adversarial, zero-sum, reproducible signal against a calibrated and world-class opponent, which lets us measure the quantities the theory predicts. Every empirical claim in this thesis is reported with sample size and confidence interval, and every “aspirational” number is labelled as a target rather than a measurement.

Chapter 2

The Branchless Algebra

2.1 Words, masks, and the uniform sublattice

Definition 2.1 (Word algebra). Fix a width $w \in \mathbb{N}$. The set of w -bit words is $\mathbb{B}^w = \{0, 1\}^w$. Equipped with bitwise conjunction $\wedge$, disjunction $\vee$, complement $\neg$, and exclusive-or $\oplus$, the structure $(\mathbb{B}^w, \wedge, \vee, \neg, 0, \mathbf{1})$ is a Boolean algebra, where $\mathbf{1} = 1^w$ is the all-ones word. Equivalently $(\mathbb{B}^w, \oplus, \wedge)$ is the Boolean ring $\mathbb{F}_2^w$.

Definition 2.2 (Uniform masks). The *uniform sublattice* is $\mathbb{U} = \{0^w, \mathbf{1}\} \subset \mathbb{B}^w$. A *mask* is any element of $\mathbb{B}^w$; a *decision mask* is an element of $\mathbb{U}$.

Lemma 2.1. $(\mathbb{U}, \wedge, \vee, \neg)$ is isomorphic to the two-element Boolean algebra $(\mathbb{B}, \wedge, \vee, \neg)$.

Proof. The map $\mathbf{1} \mapsto 1$, $0^w \mapsto 0$ is a bijection commuting with all three operations because bitwise operations on uniform words act identically on every lane, reducing to the single-bit case. $\square$

2.2 The branchless conditional

Definition 2.3 (Select). For $m \in \mathbb{U}$ and $a, b \in \mathbb{B}^w$,

$$\text{sel}(m, a, b) := (m \wedge a) \vee (\neg m \wedge b).$$

Definition 2.4 (Predicate-mask homomorphism). Let $\varphi : \mathbb{B} \rightarrow \mathbb{U}$ send $\top \mapsto \mathbf{1}$ and $\perp \mapsto 0^w$. By Lemma 2.1, φ is a Boolean-algebra isomorphism; in particular $\varphi(p \wedge q) = \varphi(p) \wedge \varphi(q)$, etc.

Theorem 2.1 (Conditional elimination). For any predicate p and values a, b ,

$$\llbracket \text{if } p \text{ then } a \text{ else } b \rrbracket = \text{sel}(\varphi(p), a, b).$$

Proof. If $p = \top$ then $\varphi(p) = \mathbf{1}$ and $\text{sel}(\mathbf{1}, a, b) = (\mathbf{1} \wedge a) \vee (0^w \wedge b) = a$. If $p = \perp$ then $\varphi(p) = 0^w$ and $\text{sel}(0^w, a, b) = 0 \vee b = b$. These exhaust the cases. $\square$

Remark 2.1. Theorem 2.1 replaces a control-flow branch by a closed-form algebraic expression. The cost of sel is data-independent: it always evaluates both a and b . The luxury that branchy code buys—skipping the unused arm—is exactly the property we will show is fatal under entropy and latency stress.

2.3 Cyclomatic complexity and the Radon condition

Definition 2.5 (Cyclomatic number). For a control-flow graph $G = (N, E)$ with P connected components, the cyclomatic number is $M(G) = |E| - |N| + 2P$. A routine R is *branchless*, written $\text{CC}(R) = 1$, iff $M(G(R)) = 1$; equivalently its CFG is a single directed path.

Theorem 2.2 (Control-to-algebra compilation). *Let R be a total decision procedure over a finite input $\mathcal{I}$ whose loops have compile-time-bounded trip counts and whose branches test predicates from a fixed finite family. Then R admits an extensionally equal implementation R' with $\text{CC}(R') = 1$ over $(\mathbb{B}^w, \wedge, \vee, \neg, \oplus, \text{sel})$.*

Proof sketch. Apply Theorem 2.1 to each branch, replacing it by a sel over both arms; the arms are themselves compiled recursively. Fully unroll each bounded loop to its maximal trip count, masking the effect of iteration i by $\varphi(i < \text{len})$ so that surplus iterations are inert. The result has no data-dependent edges, hence a single CFG path, hence $\text{CC} = 1$. Extensional equality holds branch-by-branch by Theorem 2.1 and iteration-by-iteration by the inertness of masked-off work. $\square$

Example 2.1 (Saturating distance). The data-dependent idiom *if $i > j$ then $i - j$ else $j - i$* compiles to the branchless $|i - j| = (i \oplus s) - s$ with $s = \text{sgn}(i - j)$ realized as an arithmetic-shift mask; no comparison branch survives.

Chapter 3

The Calculus of Data-Oblivious Latency

3.1 The latency functional

Definition 3.1 (Latency functional). For a routine R and machine model, the *latency functional* $\mathcal{L}_R : \mathcal{I} \rightarrow \mathbb{R}_{>0}$ assigns to each input i the number of cycles to evaluate $R(i)$. Its worst case is $\text{WCET}(R) = \sup_{i \in \mathcal{I}} \mathcal{L}_R(i)$.

Definition 3.2 (Data-obliviousness). R is *data-oblivious* iff $\mathcal{L}_R(i) = \mathcal{L}_R(i')$ for all i, i' of equal size n . We then write $\mathcal{L}_R(i) = c_R(n)$.

Proposition 3.1 (Zero-variance latency). *If R is data-oblivious then over any input distribution supported on size- n inputs, $\text{Var}_i[\mathcal{L}_R] = 0$ and $\text{WCET}(R) = c_R(n)$ is tight.*

Proof. $\mathcal{L}_R$ is constant on the support, so its variance is zero and its supremum equals its (attained) value. Tightness is immediate. $\square$

Corollary 3.1. *A $\text{CC} = 1$ routine produced by Theorem 2.2 on fixed-size inputs is data-oblivious; hence it admits a certifiable, tight WCET.*

3.2 Why branchy latency grows with entropy

Definition 3.3 (Misprediction load). Let $\mu_R(i)$ count branch mispredictions incurred by branchy R on input i , and let c_{mp} be the per-miss penalty. Then $\mathcal{L}_R(i) = \mathcal{L}_R^0(i) + c_{\text{mp}} \mu_R(i)$, where $\mathcal{L}^0$ is the mispredict-free cost.

Theorem 3.1 (Entropy lower bound on branchy cost). *Consider a data-dependent branch whose taken/not-taken outcome is a stationary process of Shannon entropy rate $H \in [0, 1]$ bits per evaluation. Any predictor with bounded state achieves an asymptotic misprediction rate $\bar{\mu} \geq g(H)$ where g is strictly increasing with $g(0) = 0$. Consequently the expected per-branch latency $\mathbb{E}[\mathcal{L}_R]$ is strictly increasing in H .*

Proof sketch. A finite-state predictor induces a coding of the outcome sequence; by Fano's inequality the achievable error rate of predicting the next symbol is bounded below by a strictly increasing function of the conditional entropy, which for a memoryless-dominated source reduces

to a strictly increasing $g(H)$ with $g(0) = 0$ (a perfectly predictable branch is free). Substituting $\bar{\mu} \geq g(H)$ into the latency identity gives the claim. $\square$

Proposition 3.2 (Volatility invariance). *Let κ be a complexity coordinate inducing branch-outcome entropy $H(\kappa)$ that is increasing in κ . For branchy R , $\partial \mathbb{E}[\mathcal{L}_R]/\partial \kappa > 0$. For data-oblivious R , $\mu_R \equiv 0$ and $\partial \mathcal{L}_R/\partial \kappa = 0$.*

Proof. Compose Theorem 3.1 with monotone $H(\kappa)$ for the branchy case; data-obliviousness gives $\mu_R \equiv 0$, hence constancy in κ . $\square$

Remark 3.1 (The economic content). Proposition 3.2 is the formal version of “fastest when it matters most.” A branchy decision engine is slowest exactly when the input is most chaotic—the market crash, the news spike—which is when latency is most valuable. A data-oblivious engine is flat across κ . Determinism is not a nicety; it is the property that survives the tail.

3.3 Skill density and the shape of search

Definition 3.4 (Strength and skill density). Let $E_A(\tau)$ be the strength (Elo) of engine A at latency budget τ . The *skill density* is $\rho_A(\tau) = \frac{dE_A}{d\tau}$.

Proposition 3.3 (Logarithmic search law). *For an adaptive search engine S with effective branching factor $b > 1$ and node rate r , search depth obeys $d(\tau) \approx \log_b(\tau r)$, and if each ply contributes $\gamma > 0$ Elo then*

$$E_S(\tau) \approx E_0 + \gamma \log_b(\tau r), \quad \rho_S(\tau) = \frac{\gamma}{\tau \ln b}.$$

Moreover S has a hard feasibility boundary $\tau_{\min}(S) > 0$, the least budget at which $d \geq 1$ completes; for $\tau < \tau_{\min}$, S returns no lawful action and E_S is undefined.

Proof sketch. Nodes searched scale as $r\tau$; with branching b the deepest fully-searched ply is $\log_b(r\tau)$. Linear-in-depth strength is the standard self-play scaling, measured here directly (Chapter 7). Differentiate for ρ . The boundary $\tau_{\min}$ is the wall-clock of one legal-move generation plus a depth-one evaluation; below it no move is emitted. $\square$

Proposition 3.4 (Branchless plateau). *A data-oblivious engine B with fixed per-decision cost τ_B satisfies $E_B(\tau) = E_B$ (constant) for $\tau \geq \tau_B$ and is undefined below, with $\tau_B \ll \tau_{\min}(S)$ whenever B omits search.*

Proof. A fixed-work evaluator produces the same action regardless of additional budget, so its strength is constant above its (constant) cost. Omitting search removes the $\log_b(r\tau)$ term and collapses τ_B to a single evaluation, strictly below the search engine’s move-generation-plus-evaluation floor. $\square$

Chapter 4

The Geometry of the Latency–Quality Manifold

4.1 Engines as curves

Definition 4.1 (Latency–quality plane). Let $\wp = \mathbb{R}_{>0} \times \mathbb{R}$ with coordinates (τ, E) . An engine A is the curve $\Gamma_A = \{(\tau, E_A(\tau)) : \tau \in \text{dom } E_A\}$. The *gap* between S and B is $g(\tau) = E_S(\tau) - E_B(\tau)$ on the common domain.

Theorem 4.1 (Inversion Theorem). *Assume Propositions 3.3 and 3.4. Then:*

- (i) *On $[\tau_{\min}(S), \infty)$ the gap g is strictly increasing: $g'(\tau) = \rho_S(\tau) = \gamma/(\tau \ln b) > 0$. Hence as the budget shrinks toward $\tau_{\min}$, the gap shrinks.*
- (ii) *On $[\tau_B, \tau_{\min}(S))$, E_S is undefined while E_B is defined; B is the unique feasible engine.*
- (iii) *If $E_B \in \text{range}(E_S)$ there is a unique parity budget τ_c with $g(\tau_c) = 0$, and $E_B \geq E_S(\tau)$ for all $\tau \leq \tau_c$.*

Proof. (i) E_B is constant, so $g' = E'_S = \rho_S > 0$ by Proposition 3.3; strict monotonicity follows. (ii) Immediate from the domains in Propositions 3.3–3.4 and $\tau_B \ll \tau_{\min}$. (iii) E_S is a continuous strictly increasing surjection onto its range, so g has a unique zero where it crosses the constant E_B ; monotonicity of g gives the inequality for $\tau \leq \tau_c$. $\square$

Corollary 4.1 (Empty category). *For every $\tau \in [\tau_B, \tau_{\min}(S))$ the comparison set contains no search engine. The sub- $\tau_{\min}$ band of $\wp$ is occupied solely by data-oblivious engines.*

4.2 The two-dimensional latency manifold

Definition 4.2 (Latency surface). Over coordinates (τ, κ) define the realized latency surface $\Lambda_A(\tau, \kappa)$ as the per-decision time of A at budget τ on inputs of complexity κ .

Proposition 4.1 (Flatness = determinism). *For data-oblivious B , $\Lambda_B(\tau, \kappa) = \tau$ is a flat plane (independent of κ). For branchy S , $\partial \Lambda_S / \partial \kappa > 0$: the surface tilts upward in complexity. The tail-latency premium of B over S is the integral of this tilt over the volatility distribution.*

Proof. Apply Proposition 3.2 pointwise in τ . $\square$

4.3 Occupancy

Definition 4.3 (Occupancy function). $\Omega(\tau) = \{\mathbf{A} : \tau \in \text{dom } E_{\mathbf{A}}\}$ is the set of engines that can act within budget τ .

Proposition 4.2. Ω is monotone: $\tau_1 \leq \tau_2 \Rightarrow \Omega(\tau_1) \subseteq \Omega(\tau_2)$, and $\lim_{\tau \downarrow 0} \Omega(\tau)$ retains only engines with $\tau_{\mathbf{A}} \rightarrow 0$, i.e. the data-oblivious ones. The lower latency axis is single-occupancy.

Proof. Larger budgets admit all engines feasible at smaller budgets, giving monotonicity. As $\tau \downarrow 0$ every engine with positive feasibility floor is excluded; only those whose floor tends to 0—the fixed-tiny-cost evaluators—survive. $\square$

Chapter 5

The Branchless Decision Engine

5.1 The evaluator as an affine form

Let $\psi : \mathcal{X} \rightarrow \mathbb{B}^m$ project a state x to a sparse feature word and let $W \in \mathbb{R}^{h \times m}$, $c \in \mathbb{R}^h$ be the first layer. The accumulator $a(x) = W\psi(x) + c$ is computed branchlessly by summing the columns of W indexed by the set bits of $\psi(x)$; iteration over set bits is data-oblivious in WCET when padded to the maximal popcount.

Lemma 5.1 (Split-ReLU signed identity). *For all $e \in \mathbb{R}$, $\text{relu}(e) - \text{relu}(-e) = e$, where $\text{relu}(t) = \max(t, 0)$ is realized branchlessly as $t \wedge \neg(t \gg (w-1))$ on the sign bit.*

Proof. If $e \geq 0$: $\text{relu}(e) = e$, $\text{relu}(-e) = 0$, difference e . If $e < 0$: $\text{relu}(e) = 0$, $\text{relu}(-e) = -e$, difference e . $\square$

Remark 5.1. Lemma 5.1 lets a rectifier network represent *signed* evaluations through a non-negative activation by pairing a neuron with its mirror; the value head $\sum_h \text{relu}(a_h) v_h$ with $v = (+1, -1, 0, \dots)$ and $a_1 = -a_0$ returns the signed score a_0 with no branch. This is the device by which the engine’s GPU value head computes a signed material-positional evaluation.

5.2 Divergence-free batched evaluation

Because the evaluator is data-oblivious, N states evaluate in lockstep with no lane divergence; on a 40-core single-instruction-multiple-data device this realizes throughput $\Theta(N)$ per dispatch. Empirically the kernel sustains 1.77×10^7 evaluations per second standalone. Search that consumes the kernel one state at a time is dispatch-latency bound; leaf-parallel batching with virtual loss restores $\sim 1.5 \times 10^6$ evaluations per second *inside* the search, recovering two orders of magnitude.

5.3 Two search operators

We instantiate two adaptive operators on top of the branchless evaluator: (a) a negamax alpha-beta search with quiescence, a transposition table, and reductions; and (b) a Monte-Carlo tree search with PUCT selection and the leaf-parallel batching above. Both are branchy by construction—Proposition 3.3 applies to them—and serve to populate the $\tau \geq \tau_{\min}$ portion of

the curve. The *branchless* portion is the depth-one evaluator alone, which occupies the sub- $\tau_{\min}$ band of Corollary 4.1.

Chapter 6

Branchless Process Conformance

6.1 Token replay

Definition 6.1 (Marked Petri net). A net is $\mathcal{N} = (P, T, F, m_0)$ with places P , transitions T , flow F , and initial marking $m_0 \in \mathbb{B}^{|P|}$ encoded as a bit per place. A transition with input mask ι and output mask o fires on marking m by

$$m' = (m \setminus \iota) \vee o, \quad \text{missing} += \text{pc}(\iota \wedge \neg m), \quad \text{consumed} += \text{pc}(\iota), \quad \text{produced} += \text{pc}(o),$$

all data-oblivious.

Definition 6.2 (Conformance fitness). $\text{fit} = 1 - \frac{\text{missing} + \text{remaining}}{\text{consumed} + \text{missing} + \text{produced}}$, with $\text{fit} = 1$ when the denominator is zero. A run is *lawful* iff $\text{fit} = 1$ and no tokens remain off the idle place.

Proposition 6.1 (Auditable decisions). *The decision pipeline $\text{Ready} \rightarrow \text{Perceive} \rightarrow \text{Evaluate} \rightarrow \text{Search} \rightarrow \text{Select} \rightarrow \text{Commit} \rightarrow \text{Ready}$ is a net $\mathcal{N}_{\text{dec}}$; replaying an engine’s per-stage event log through $\mathcal{N}_{\text{dec}}$ yields $\text{fit} = 1$ for lawful executions and $\text{fit} < 1$ for any skipped or reordered stage. Because replay is data-oblivious, the audit itself has a tight WCET.*

Remark 6.1. This is the compliance content: each decision carries a hash-chained, conformance-checked receipt proving it followed the lawful process—no skipped risk check, no reordered stage—and the proof is constant-time. A negative control (deliberately omitting the **Evaluate** stage) is rejected with $\text{fit} = 0.983 < 1$.

Chapter 7

Chess as Instrument: Empirical Results

7.1 Apparatus

The engine of Chapter 5 plays a calibrated, world-class opponent (Stockfish) under an independent arbiter that adjudicates legality, mate, stalemate, and draw. Games are seeded from a curated, balanced opening book (52 lines) played with both colours, yielding 104 independent games per condition; this defeats the determinism of the engines (identical starts give identical games). Each result is a score rate $\sigma \in [0, 1]$ with a Wald 95% interval.

7.2 The $E(\tau)$ curve

Table 7.1 reports our engine across four latency bands against Stockfish constrained to its *minimum* search (depth one, $\approx 58 \mu\text{s}$). Elo is anchored at $E_S(\text{depth } 1) = 1500$ via $E = E_{\text{anchor}} + 400 \log_{10}(\sigma/(1 - \sigma))$.

Band (τ)	Config	Score (104 games)	95% CI	E (anchor 1500)
$\sim 3 \mu\text{s}$	fixed depth 1	7.7%	2.6–12.8%	≈ 1068
$\sim 49 \mu\text{s}$	fixed depth 2	26.0%	17.5–34.4%	≈ 1318
1 ms	iterative	33.7%	24.6–42.7%	≈ 1383
10 ms	iterative	50.0%	40.4–59.6%	≈ 1500

Table 7.1: Realized $E(\tau)$ against minimal-time Stockfish. The curve is monotone in τ and reaches *parity* ($\sigma = 0.50$, $g = 0$) at 10 ms, empirically locating the parity budget τ_c of Theorem 4.1(iii).

At the two microsecond bands our engine consumes *less total compute* than the opponent (0.31 s vs 1.34 s at $3 \mu\text{s}$; 0.87 s vs 1.43 s at $49 \mu\text{s}$) while still scoring, realizing the sub-floor regime of Corollary 4.1: below $\approx 58 \mu\text{s}$ the opponent cannot move and the band is single-occupancy (Proposition 4.2).

7.3 Per-ply strength and the search law

A self-play ladder (depth d vs $d+1$) measures the marginal Elo per ply: +58, +221, +89 Elo for $d = 1, 2, 3$, confirming the linear-in-depth premise of Proposition 3.3 (with sampling noise on individual rungs).

7.4 Equal-time speed chess

Against *full-strength* time-limited Stockfish at an identical per-move budget, the gap-narrowing of Theorem 4.1(i) appears directly: score rate rises as the shared budget falls—1.0% at 10 ms, 2.1% at 5 ms, 4.2% at 1 ms—and individual games are won outright at the millisecond budget with think-time within 10% of the opponent's.

7.5 Construction cost

The entire apparatus was constructed in a single working session of a few hours, against an opponent representing more than seventeen years of dedicated engineering—a construction-effort ratio in excess of $10^4:1$ —supporting the thesis that the sub-floor regime is reachable precisely because it requires *less* machinery, not more.

Chapter 8

Decision-Engine Elo: A Latency-Parameterized Rating

Remark 8.1 (The gap in practice). Trading engines are evaluated by profit-and-loss, Sharpe ratio, and drawdown—all absolute, regime-dependent, capital-confounded, and non-comparable across firms or epochs. No standardized, head-to-head, relative skill rating exists. Chess Elo is exactly such a rating; its trading analogue is, to our knowledge, absent.

Definition 8.1 (Decision-Engine Elo). For two engines in a shared zero-sum environment with a common latency budget τ , let $\sigma_{AA'}(\tau)$ be the expected score of A vs A'. Decision-Engine Elo is the latency-parameterized rating $E_A(\tau)$ satisfying the logistic law

$$\sigma_{AA'}(\tau) = \frac{1}{1 + 10^{-(E_A(\tau) - E_{A'}(\tau))/400}},$$

fitted from round-robin head-to-head play at each τ and anchored once.

Axiom 8.1 (Latency primacy). Skill is meaningless without a budget: an engine's rating is the curve $\tau \mapsto E_A(\tau)$, not a scalar.

Proposition 8.1 (Ranking inversion). *Two engines whose curves cross at τ_c swap rank across τ_c . Hence a scalar rating is not merely lossy but order-violating; only the curve is well-defined.*

Proof. Crossing curves give opposite signs of $E_A - E_{A'}$ on the two sides of τ_c ; the logistic law then assigns the majority of expected score to different engines on each side. $\square$

Remark 8.2. By Corollary 4.1 and Proposition 4.2, the low- τ region of the Decision-Engine Elo chart is single-occupancy by construction. Whoever defines the benchmark on which only one architecture can compete owns the category.

Chapter 9

Projection onto Ultra-High-Frequency Trading

The chess board is replaced by a limit-order book; “checkmate” by realized profit-and-loss extracted from a counterparty; the move budget by the quote-lifetime deadline. The theory transfers verbatim:

1. **Volatility-invariant latency** (Proposition 3.2): the engine’s decision latency does not increase when the market’s entropy spikes—the regime in which branchy competitors mispredict hardest.
2. **Provable WCET** (Proposition 3.1): a certifiable bound “every decision within $c(n)$ cycles, for any input,” impossible for data-dependent systems.
3. **Auditable execution** (Chapter 6): each decision ships a constant-time conformance receipt.
4. **Empty category** (Corollary 4.1): in the microsecond band, the adaptive “grandmaster” cannot act; the data-oblivious engine is the only player on the board.

The product is not a fast decision; it is a *constant-time, certifiable, auditable* decision whose competitive advantage *grows* as the deadline tightens into the band where the business lives.

Chapter 10

Conclusion

We have shown that data-dependent control flow is dispensable (Chapter 2), that dispensing with it yields zero-variance, certifiable latency (Chapter 3), that the latency–quality plane has a geometry in which a constant-time engine necessarily and uniquely occupies the sub-feasibility band of any search engine (Chapter 4, Theorem 4.1), and that a chess instrument realizes the predicted curve to parity at 10 ms and coherent play at $3\mu\text{s}$ on a quarter of the opponent’s compute (Chapter 7). We introduced Decision-Engine Elo as the rating the category lacks, and projected the apparatus onto UHFT, where the inversion is not a curiosity but the entire value proposition.

Anticipated defense questions

1. *Is depth-one Stockfish a fair opponent?* It is the opponent’s *minimum-time* configuration; by Axiom 8 the relevant comparison is at a fixed budget, and below $\tau_{\min}$ no fairer opponent exists because none exists at all (Corollary 4.1).
2. *The engine is weak at long time controls.* Granted, and irrelevant: Theorem 4.1 concerns the short-budget regime, and Proposition 4.2 shows that regime is where the architecture is unique.
3. *Are the branchless primitives novel?* Many are classical (mask-select, split-ReLU, SWAR); the contribution is the *synthesis*—a constant-time, certifiable decision kernel—and the *Inversion Theorem* with its empirical realization, not the individual gadgets.
4. *Sample size?* Each curve point is 104 independent games with reported confidence intervals; the methodology scales to arbitrarily many via book expansion.

Appendix A

Notation

Symbol	Meaning
$\mathbb{B}^w$	w -bit words; Boolean algebra / ring $\mathbb{F}_2^w$
$\mathbb{U} = \{0^w, \mathbf{1}\}$	uniform (decision) masks; $\cong \mathbb{B}$ (Lemma 2.1)
$\text{sel}(m, a, b)$	branchless select $(m \wedge a) \vee (\neg m \wedge b)$
$\varphi : \mathbb{B} \rightarrow \mathbb{U}$	predicate-mask isomorphism
$\text{CC}(R)$	cyclomatic number of routine R ; branchless $\Leftrightarrow 1$
$\mathcal{L}_R(i)$	latency functional (cycles) of R on input i
$\text{WCET}(R)$	worst-case execution time $\sup_i \mathcal{L}_R(i)$
$\mu_R(i)$	branch mispredictions; c_{mp} per-miss penalty
H	Shannon entropy (rate) of a branch outcome
κ	complexity / volatility coordinate
τ	latency budget per decision
$E_A(\tau)$	strength (Elo) of engine A at budget τ
$\rho_A = dE/d\tau$	skill density
b, r, γ	branching factor, node rate, Elo-per-ply
$\tau_{\min}(\mathbf{S})$	feasibility floor of search engine $\mathbf{S}$
$g(\tau) = E_S - E_B$	quality gap
τ_c	parity budget, $g(\tau_c) = 0$
$\Omega(\tau)$	occupancy: engines feasible within τ
$\Lambda_A(\tau, \kappa)$	realized latency surface
relu, pc	rectifier, popcount (both branchless)
fit	Petri-replay conformance fitness
σ	head-to-head score rate in $[0, 1]$

Appendix B

The Logistic and Performance-Rating Identities

From the logistic law, the performance rating of A against an anchor of rating R_0 given score σ over n games is

$$E_A = R_0 + 400 \log_{10} \frac{\sigma}{1 - \sigma},$$

with Wald interval on σ of half-width $1.96\sqrt{\sigma(1 - \sigma)/n}$ propagated through the (monotone) map above. The per-ply ladder uses the same identity with the opponent being the engine itself at one greater depth, isolating γ of Proposition 3.3.